

# claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

## Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a507edff5953ba0ca.jsonl`  
- SHA-256: `e282c7b8084ad33aa4f46b18f712ab422fba0d65324661cdfa917d1250a92c32`  
- Source modified: `2026-06-16T03:27:26+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

## Transcript

1. user (2026-06-16T03:25:50.176Z)

### Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

### Research question

Research question: **\*\*What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?\*\*** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **\*recent\*** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

## Claim under review

"Reframing boundary regression as learning a probability distribution over discrete temporal bins (GFL-inspired Boundary Distribution Regression head) directly targets the 'fuzzy'/ambiguous boundary problem where forcing a single deterministic offset gives a noisy, unstable training signal ? the same conflation that causes over-merged segments. This is the mechanism most relevant to splitting/refining merged windows."

**\*\*Source:\*\*** <https://arxiv.org/pdf/2512.01298> (primary)

**\*\*Supporting quote:\*\*** "Inspired by Generalized Focal Loss (GFL) [6], we reformulate this deterministic regression problem as a probabilistic one. Instead of predicting a single value ds, our BDR head predicts a discrete probability... However, in reality, action boundaries are often fuzzy. Forcing the model to regress to a single point in such cases provides a noisy and unstable learning signal [4]."

## Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

**\*\*refuted=true\*\*** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

**\*\*refuted=false\*\*** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:25:52.592Z)

I'll verify this claim about the Boundary Distribution Regression (BDR) head. Let me start by examining the primary source.

3. assistant (2026-06-16T03:25:55.329Z)

I'll fetch the primary source to verify the supporting quote and the claim's interpretation.

4. user (2026-06-16T03:26:04.634Z)

Analysis of PDF Content

Paper Details

- \*\*Title\*\*: TBT-Former: Learning Temporal Boundary Distributions for Action Localization
- \*\*Task\*\*: Temporal action localization in videos
- \*\*Domain\*\*: Computer vision (cs.CV)
- \*\*Publication Date\*\*: December 2, 2025 (arXiv)
- \*\*Authors\*\*: Thisara Rathnayaka, Uthayasanker Thayasivam

Key Findings

Boundary Distribution Regression (BDR) Discussion

The PDF content is heavily compressed and primarily contains PDF structural data rather than readable text. However, the title itself indicates the paper's core contribution: learning temporal boundary **distributions** rather than point estimates.

Stated Motivation

Based on the available metadata and title structure, the paper's motivation appears to center on:

- Reformulating boundary localization from deterministic point regression to **probabilistic distribution modeling**
- Addressing instability in action boundary localization through distribution-based approaches

Critical Quote Issues

The document's compressed nature prevents extraction of exact quotes about:

- Explicit discussion of "fuzzy" boundaries forcing "noisy and unstable learning signals"
- Specific GFL (Generalized Focal Loss) comparisons
- Explicit over-segmentation or merged segment fixes

The term "distribution" in the title suggests the paper treats boundaries as probabilistic rather than deterministic, but the actual textual content supporting this cannot be reliably extracted from the provided PDF source.

[Binary content (application/pdf, 698.3KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579953015-t4vlj4.pdf]

5. user (2026-06-16T03:26:09.302Z)

Web search results for query: "arxiv 2512.01298 Boundary Distribution Regression Generalized Focal Loss temporal action"

Links: [{"title": "[2512.01298] TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/abs/2512.01298"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://www.arxiv.org/pdf/2512.01298"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/html/2512.01298"}, {"title": "Boundary Uncertainty in a Single-Stage Temporal Action Localization Network", "url": "https://arxiv.org/pdf/2008.11170"}, {"title": "[1705.01180] Cascaded Boundary Regression for Temporal Action"}]

Detection", "url": "https://arxiv.org/abs/1705.01180"}, {"title": "[2011.12885] Generalized Focal Loss V2: Learning Reliable Localization Quality Estimation for Dense Object Detection", "url": "https://arxiv.org/abs/2011.12885"}, {"title": "Generalized Focal Loss: Learning Qualified and Distributed Bounding Boxes for", "url": "https://proceedings.neurips.cc/paper/2020/file/f0bda020d2470f2e74990a07a607ebd9-Paper.pdf"}, {"title": "[2006.04388] Generalized Focal Loss: Learning Qualified and Distributed Bounding Boxes for Dense Object Detection", "url": "https://arxiv.org/abs/2006.04388"}, {"title": "[2006.04388] Generalized Focal Loss: Learning Qualified and Distributed Bounding Boxes for Dense Object Detection", "url": "https://ar5iv.labs.arxiv.org/html/2006.04388"}]

Based on the search results, I found the paper you're looking for:

## TBT-Former: Learning Temporal Boundary Distributions for Action Localization

The paper introduces the Temporal Boundary Transformer (TBT-Former), which enhances the ActionFormer baseline with three core contributions: a higher-capacity scaled Transformer backbone, a cross-scale feature pyramid network, and a novel boundary distribution regression head inspired by the principles of Generalized Focal Loss that recasts boundary regression as a probability distribution learning problem.

The paper addresses Temporal Action Localization (TAL), which aims to identify the start time, end time, and category of all action instances within untrimmed videos.

The Boundary Distribution Regression (BDR) head learns a flexible probability distribution over a range of possible temporal locations instead of regressing a single offset value, allowing the model to explicitly represent boundary uncertainty for more robust and accurate localization, especially for actions with ambiguous transitions.

Through extensive experimentation, TBT-Former demonstrates superior performance, setting state-of-the-art on the THUMOS14 and EPIC-Kitchens 100 datasets and achieving highly competitive results on ActivityNet-1.3.

You can access the paper directly at: <https://arxiv.org/abs/2512.01298>

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 6. assistant (2026-06-16T03:26:11.744Z)

The PDF text wasn't fully extracted. Let me get the HTML version to verify the exact quote and the stated motivation precisely.

### 7. user (2026-06-16T03:26:21.436Z)

## Analysis of BDR Head Passages

### (1) "Noisy and Unstable Learning Signal"

**Exact quote:**

"Forcing the model to regress to a single point in such cases provides a noisy and unstable learning signal"

This appears in Section IV-D, Motivation subsection.

### (2) Purpose/Problem Statement

The BDR head addresses **boundary uncertainty and ambiguous transitions**, NOT over-segmentation or segment merging.

**Exact quote on purpose:**

"action boundaries are often fuzzy. Forcing the model to regress to a single point in such cases

provides a noisy and unstable learning signal"

The stated problem is imprecise localization of actions with "ambiguous or 'fuzzy' temporal boundaries."

### (3) Over-segmentation/Merge/Split References

**\*\*Result:\*\*** The paper contains **\*\*NO mentions\*\*** of "over-segment," "merge," "split," or "fragment" in connection with the BDR head or boundary problems.

The focus is entirely on boundary ambiguity/uncertainty, not on segment fragmentation or merging issues.

### (4) Reference Identifications

- **\*\*Reference [4]:\*\*** Shi et al., "TriDet: Temporal Action Detection with Relative Boundary Modeling" (CVPR 2023)

- **\*\*Reference [6]:\*\*** Li et al., "Generalized Focal Loss: Learning Qualified and Distributed Bounding Boxes for Dense Object Detection" (NeurIPS 2020)

### 8. assistant (2026-06-16T03:26:24.892Z)

I now have the key facts. Let me verify the broader claim about whether GFL-style distributional regression is "the mechanism most relevant to splitting/refining merged windows" by checking whether the literature connects boundary uncertainty modeling to over-segmentation control.

### 9. user (2026-06-16T03:26:41.546Z)

Web search results for query: "over-segmentation temporal action localization boundary regression versus action segmentation MS-TCN ASRF smoothing loss"

Links: [{"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}, {"title": "Combining Boundary Supervision and Segment-Level Regularization for Fine-Grained Action Segmentation", "url": "https://arxiv.org/html/2604.01859"}, {"title": "Boundary-Centric Active Learning for Temporal Action Segmentation", "url": "https://arxiv.org/html/2604.15173v1"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "Exploring Ordinal Bias in Action Recognition for Instructional Videos", "url": "https://arxiv.org/pdf/2504.06580"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | DeepAI", "url": "https://deepai.org/publication/alleviating-over-segmentation-errors-by-detecting-action-boundaries"}, {"title": "MSBATN: Multi-Stage Boundary-Aware Transformer Network for Action Segmentation in Untrimmed Surgical Videos", "url": "https://arxiv.org/html/2504.18756"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request PDF", "url": "https://www.researchgate.net/publication/342944719\_Alleviating\_Over-segmentation\_Errors\_by\_Detecting\_Action\_Boundaries"}, {"title": "Combining Boundary Supervision and Segment-Level Regularization for Fine-Grained Action Segmentation", "url": "https://arxiv.org/pdf/2604.01859"}, {"title": "Maximization and restoration: Action segmentation through dilation passing and temporal reconstruction - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/abs/pii/S003132032200245X"}]

Based on the search results, here's what I found on your topic:

## Over-Segmentation in Temporal Action Localization

Over-segmentation (fragmentation into short sub-segments) is identified as a main obstacle to accurate segmentation in long videos, motivating architectural components and losses that directly target boundaries including boundary regression and boundary-smoothing refinements.

## Key Approaches

#### **\*\*Smoothing Losses in MS-TCN:\*\***

MS-TCN proposes a model composed of stacked dilated convolutions and a smoothing loss designed to alleviate over-segmentation issues. More specifically, smoothing losses (truncated MSE on log probabilities) enforce temporal coherence, addressing the over-segmentation problem.

#### **\*\*Boundary Regression in ASRF:\*\***

ASRF proposes a framework that divides temporal action segmentation into frame-wise action classification and action boundary regression, proposing a loss function for smoothing the transition of action probabilities. ASRF improves segmentation quality by adding an auxiliary branch that explicitly regresses action boundaries to mitigate over-segmentation errors.

#### **\*\*Hybrid Approach:\*\***

Recent work combines boundary supervision with segment-level regularization. Frame-wise classification losses such as cross-entropy often lead to over-segmentation and boundary misalignment, but simple training-time enhancements augment existing models with two auxiliary losses and only one additional boundary output channel.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

#### **10. assistant (2026-06-16T03:26:45.321Z)**

This is the decisive distinction. The literature consistently attributes over-segmentation control to a different mechanism ? smoothing losses (MS-TCN) and boundary detection branches (ASRF) ? than to GFL-style distributional boundary regression. Let me confirm the precise framing of the claim under review and what GFL's distribution focal loss actually targets.

#### **11. user (2026-06-16T03:26:51.930Z)**

## **Purpose of Distributional Representation in Generalized Focal Loss**

Based on the abstract, the answer is **\*\*(a) handling ambiguous/flexible boundary distributions and improving localization accuracy\*\***.

The paper identifies a key limitation with existing approaches: "the inflexible Dirac delta distribution for localization when there is ambiguity and uncertainty in complex scenes."

The distributional approach addresses this by using "a vector to represent arbitrary distribution of box locations" rather than a single deterministic value. This allows the model to "accurately depict the flexible distribution in real data," accommodating scenarios where object boundaries may be inherently uncertain or ambiguous rather than fixed at precise pixel locations.

The motivation is fundamentally about **modeling uncertainty and complexity** in real-world detection scenarios, not about controlling over-segmentation or detection merging. The authors argue this representation "eliminate[s] the inconsistency risk" between training and inference while better reflecting the actual variability in ground-truth localization.

#### **12. assistant (2026-06-16T03:27:06.817Z)**

I now have sufficient evidence to render a verdict. Let me assess the claim against the gathered evidence.

The claim has two parts:

**\*\*Part 1 (mechanism description):\*\*** "GFL-inspired BDR reframes boundary regression as a probability distribution over discrete bins, targeting the fuzzy/ambiguous boundary problem where forcing a single deterministic offset gives a noisy, unstable training signal." This part IS well-supported. The quote is verbatim (verified in the HTML source), GFL's DFL motivation is identical (modeling ambiguous/uncertain boundaries via distributions instead of Dirac delta), and the paper's BDR head genuinely implements this.

**\*\*Part 2 (the load-bearing claim):\*\*** That this is "the same conflation that causes over-merged

segments" and "the mechanism most relevant to splitting/refining merged windows." This part is REFUTED:

1. **\*\*Overreach beyond the quote.\*\*** The quote talks about "fuzzy" boundary *\*localization\** (precise placement of a boundary that exists), NOT about whether two adjacent actions get *\*merged into one\**. These are different failure modes. The paper's stated purpose (verified) is "boundary uncertainty and ambiguous transitions" / "more robust and accurate localization" ? it never mentions over-segmentation, merging, splitting, or fragmentation anywhere in connection with the BDR head.
2. **\*\*Contradicted by the canonical literature.\*\*** Over-segmentation/merging is a distinct, well-studied problem with its OWN mechanisms: MS-TCN's smoothing loss (truncated MSE on log-probabilities) and ASRF's dedicated action-boundary *\*detection\** branch ("Alleviating Over-segmentation Errors by Detecting Action Boundaries"). The user's actual problem is the inverse of over-segmentation ? it is UNDER-segmentation (mega-windows that fail to split). The mechanism most relevant to *\*splitting merged windows\** is boundary *\*detection/regression that inserts boundaries\** (ASRF-style) plus actionness/inactivity gating ? not making an existing boundary's offset probabilistic. A distribution head refines WHERE a predicted boundary sits; it does not decide that a new boundary SHOULD exist mid-window.
3. **\*\*GFL/DFL's own purpose is localization-quality, not segment-count control\*\*** (verified from the source NeurIPS paper). It models the flexible distribution of a single box edge under ambiguity; it has no notion of merging/splitting object instances.

So the claim's framing ? that distributional boundary regression is "the mechanism most relevant to splitting/refining merged windows" ? is an unsupported inferential leap that conflates boundary *\*localization\** uncertainty with segment *\*merging\**. Two of three checklist failure conditions are met (unsupported by quote / overreach, and contradicted by credible sources).

### 13. user (2026-06-16T03:27:18.764Z)

Structured output provided successfully

### 14. assistant (2026-06-16T03:27:26.340Z)

Verdict rendered: **\*\*refuted=true\*\*** (high confidence).

The claim is half-right but its load-bearing assertion is an overreach. The first part ? that TBT-Former's GFL-inspired BDR head reframes boundary regression as a distribution over discrete bins to handle "fuzzy" boundaries (verbatim quote confirmed at [arxiv.org/html/2512.01298](https://arxiv.org/html/2512.01298), and consistent with GFL's own Dirac-delta motivation at [arXiv:2006.04388](https://arxiv.org/abs/2006.04388)) ? is accurate. But the second part, that this is "the same conflation that causes over-merged segments" and "the mechanism most relevant to splitting/refining merged windows," is unsupported by the quote and contradicted by the canonical literature:

- The paper's verified motivation for the BDR head is **\*\*boundary localization uncertainty / ambiguous transitions\*\*** ? it never mentions over-segmentation, merging, splitting, or fragmentation.
- Boundary *\*localization\** uncertainty (where an existing boundary sits) is a different failure mode from segment *\*merging\** (whether a boundary should exist mid-window).
- Over-segmentation/merging control is attributed in the literature to **\*\*distinct\*\*** mechanisms ? MS-TCN's smoothing loss and especially **\*\*ASRF's dedicated boundary-detection branch\*\*** ("Alleviating Over-segmentation Errors by Detecting Action Boundaries").
- The user's actual bug is *\*under\*-segmentation* (mega-windows), whose fix is boundary detection/inactivity gating that creates new splits ? not making an existing offset probabilistic.

Two of three checklist failure conditions are met (unsupported-by-quote/overreach + contradicted by credible primary sources).